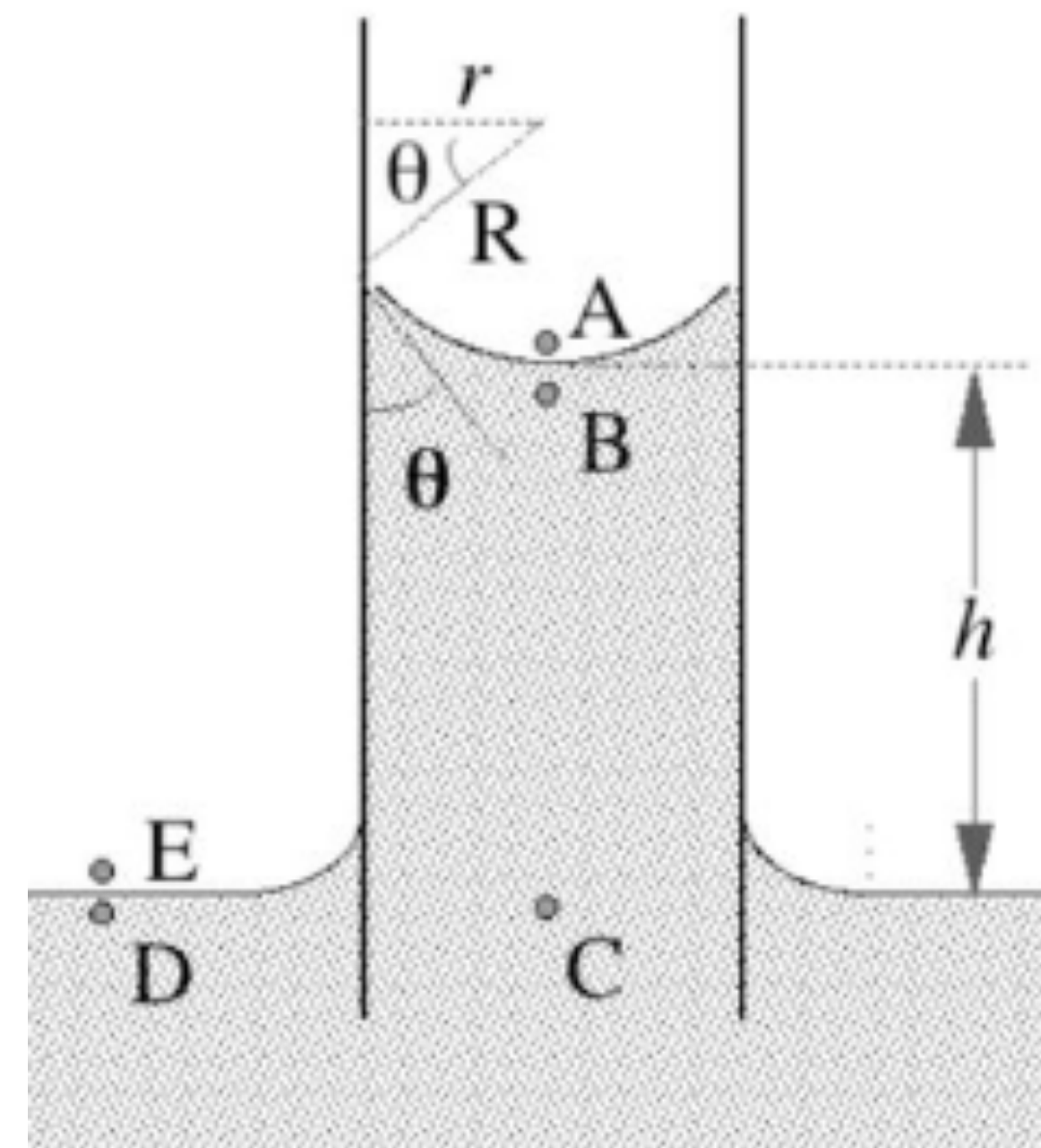


F3. Phénomènes interfaciaux impliquant des fluides

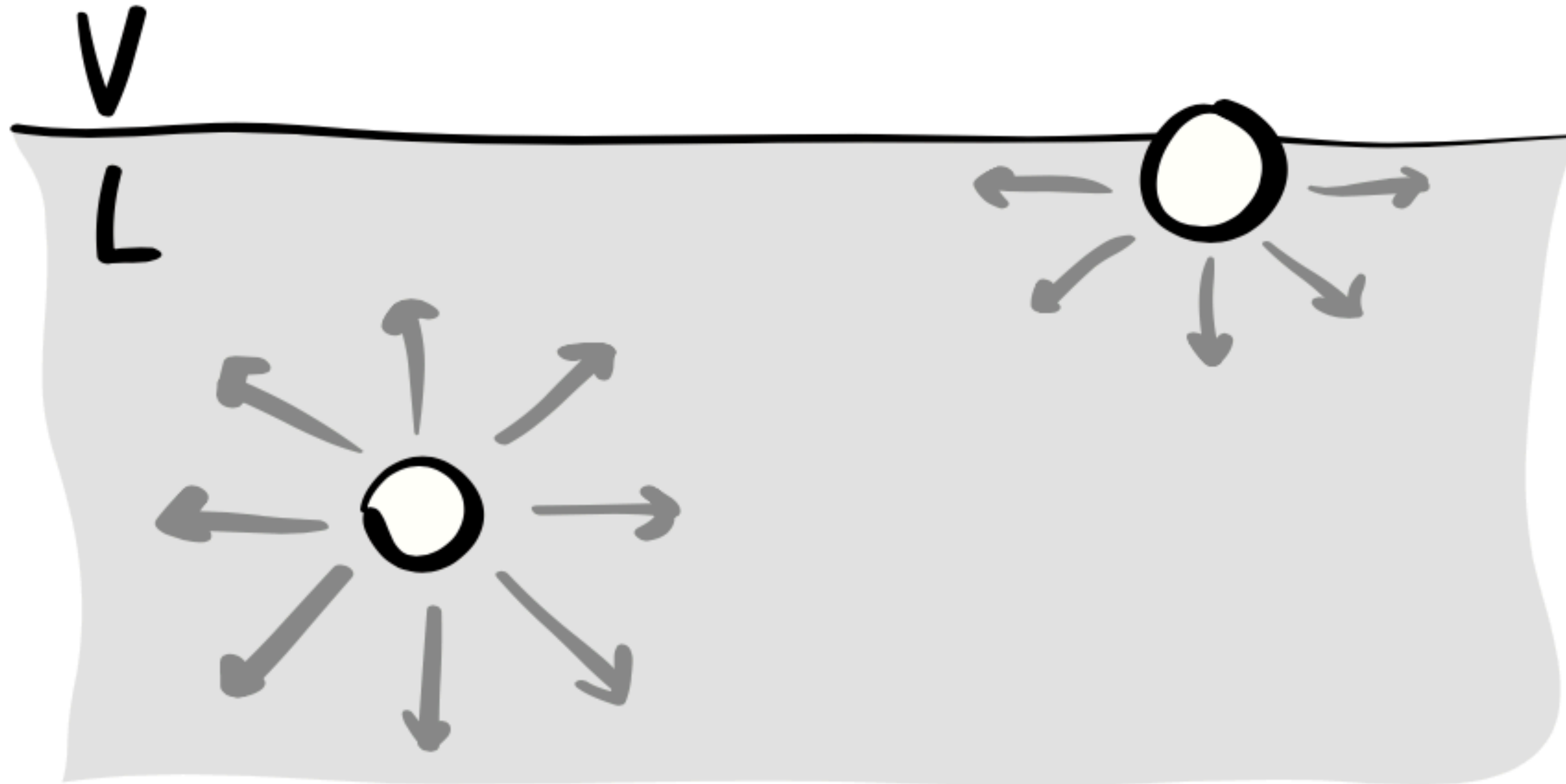
Anass Najlaoui

26/03/2026

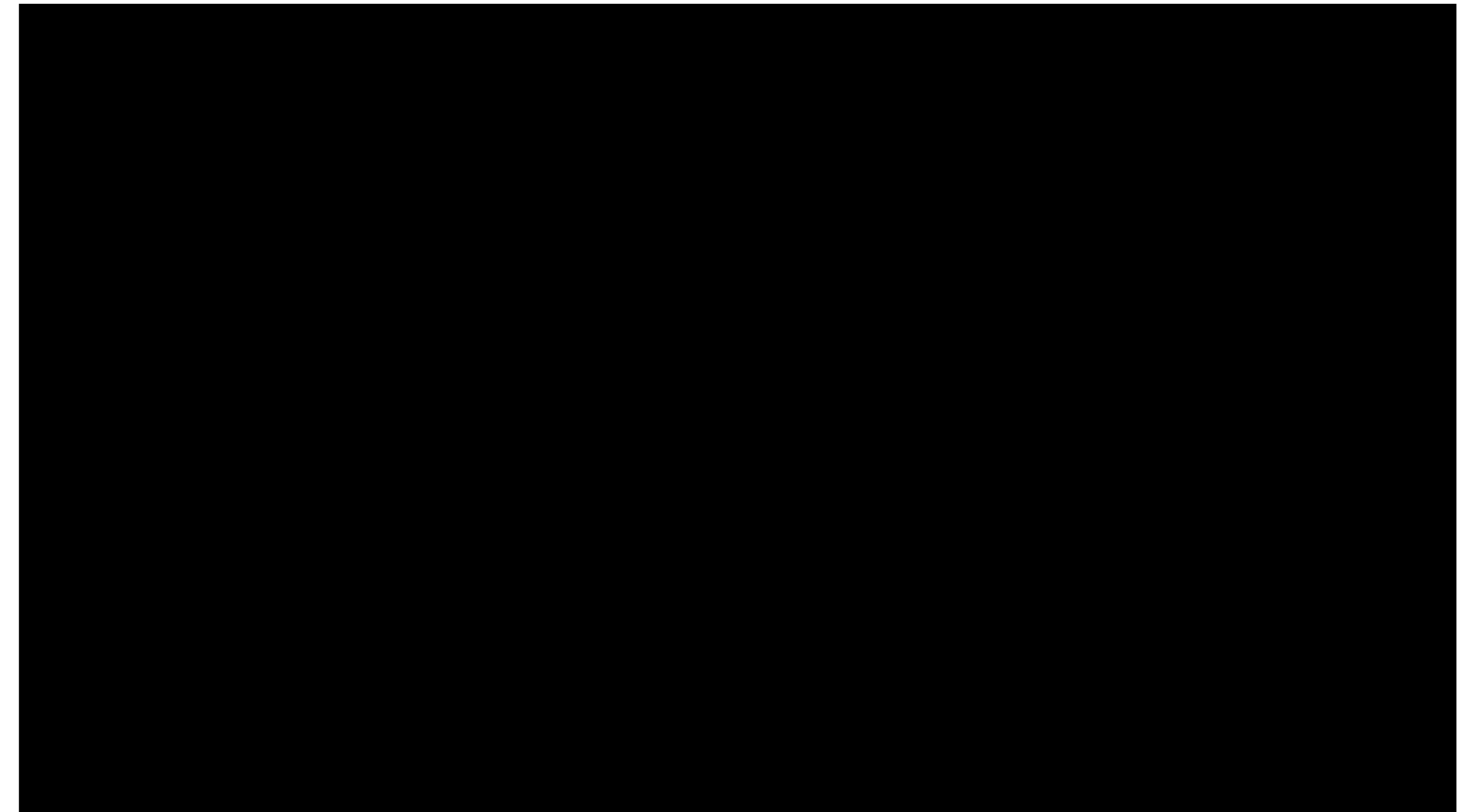
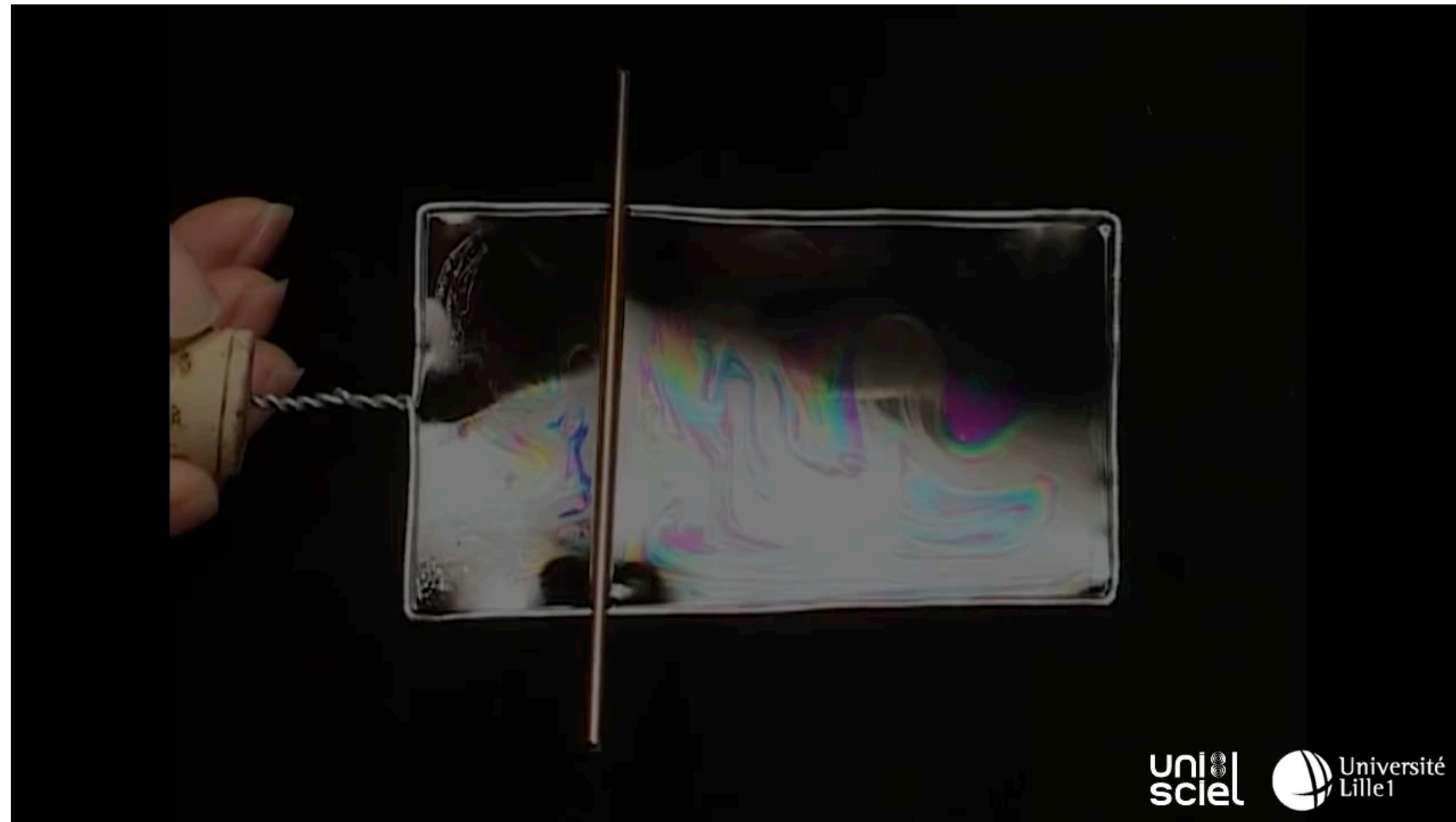


I) Tension de surface

a) Origine microscopique

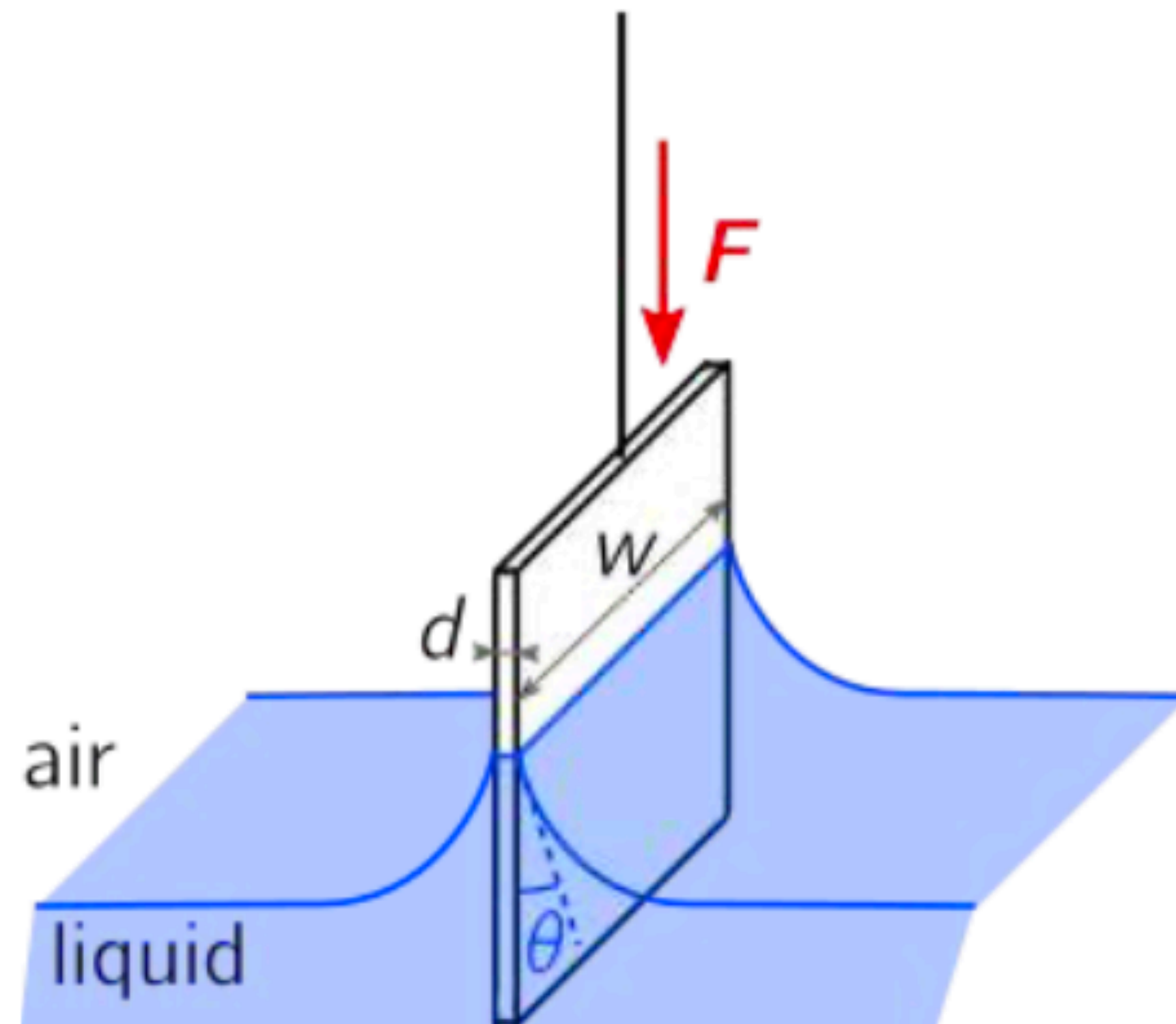


Phénomène de tension superficielle



c) Méthode de mesure

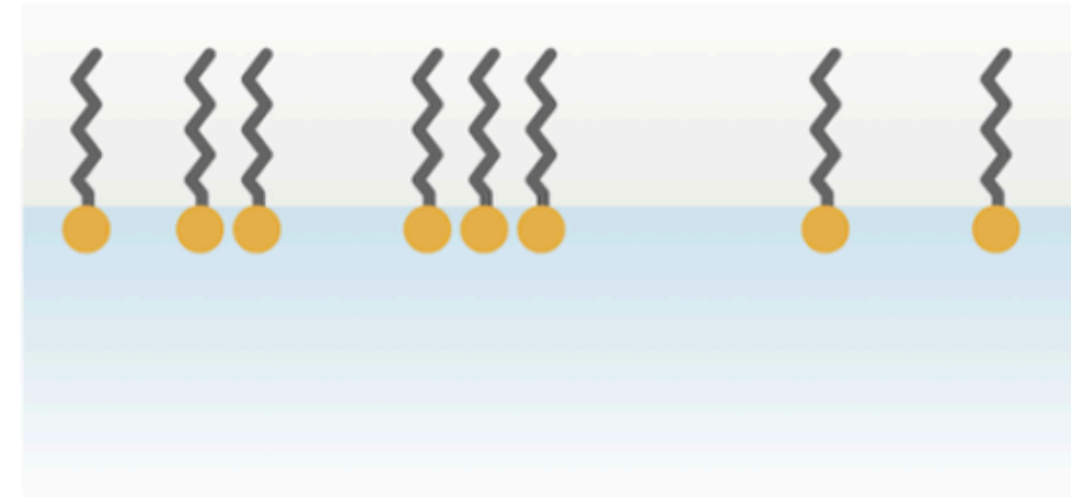
Méthode de Wilhelmy



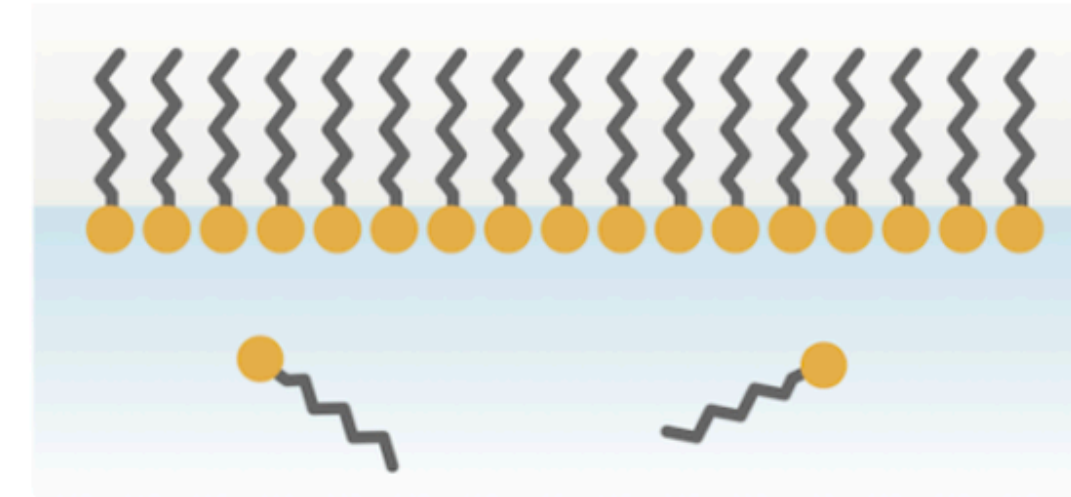
$$\gamma = F / (W \cos(\theta)).$$

Tensioactifs

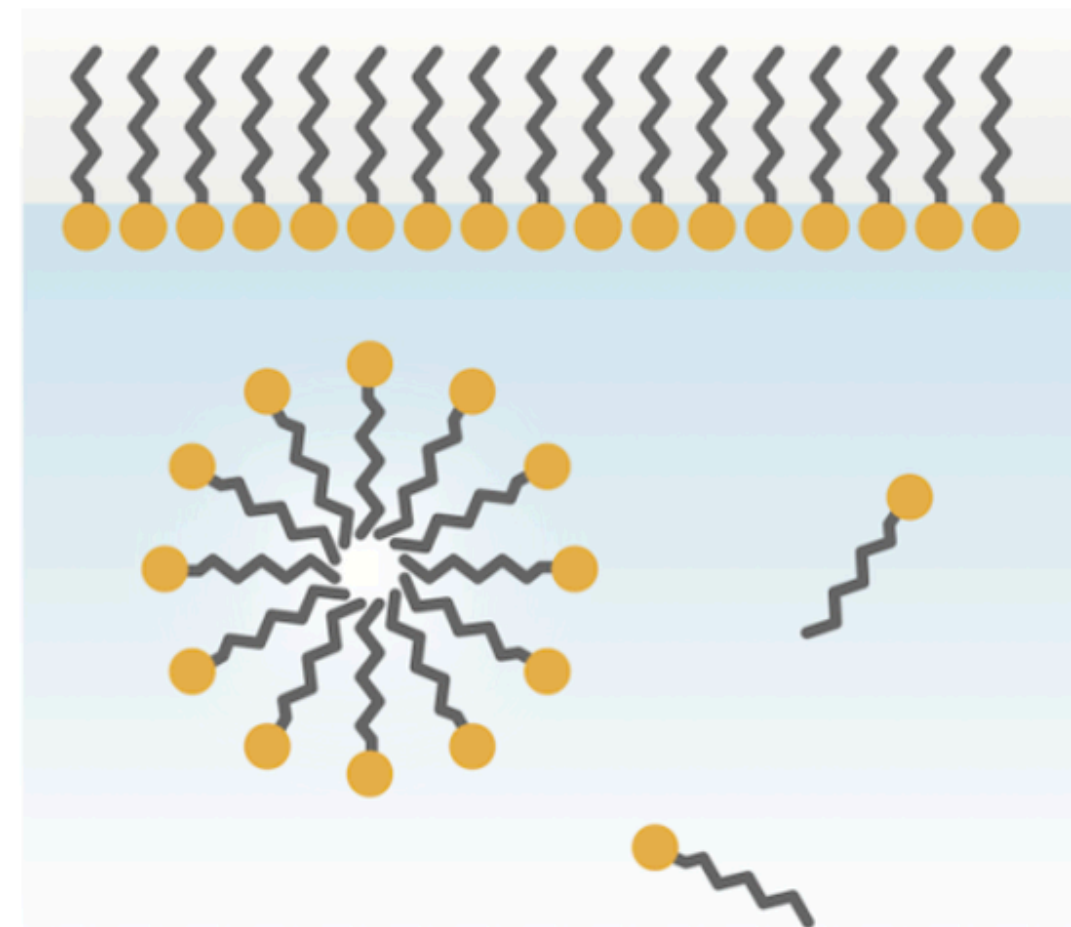
Tension de surface faible :
 $c \ll CMC$



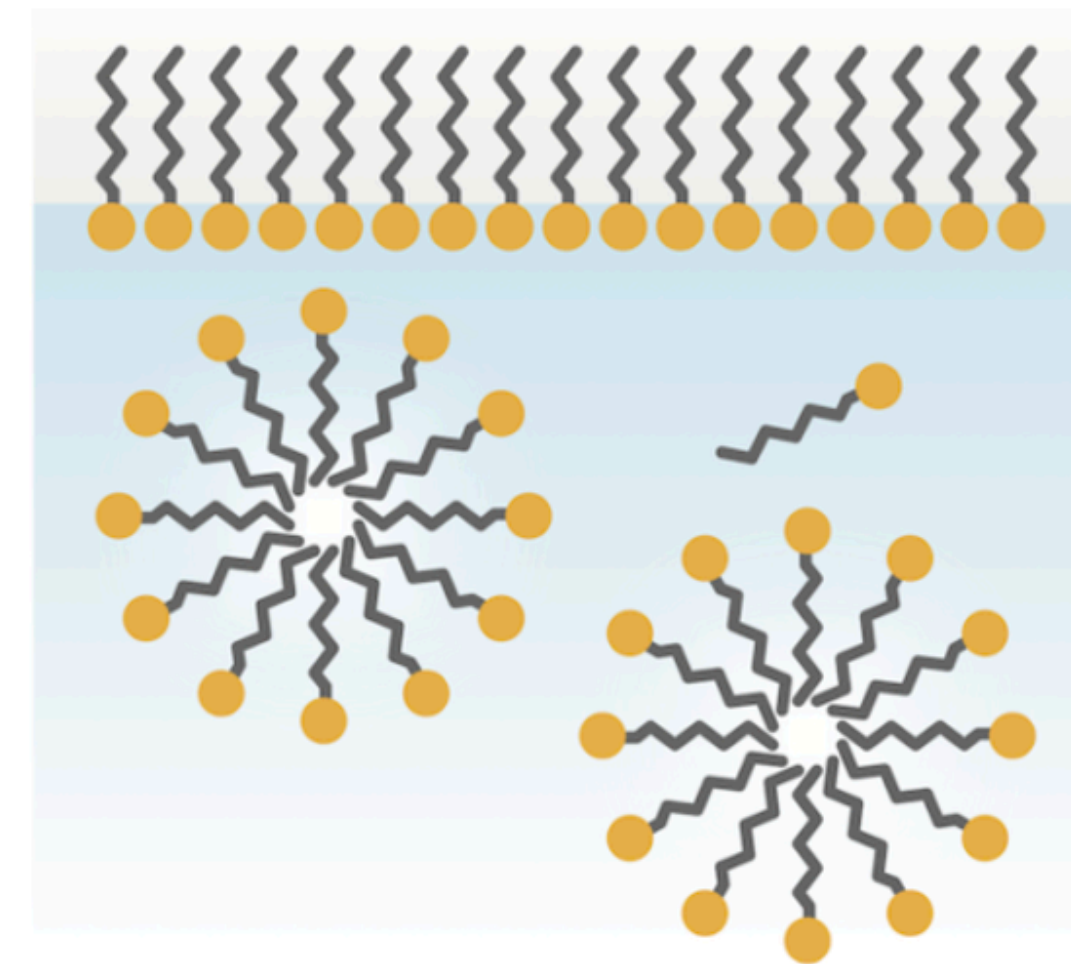
Tension de surface modérée : $c < CMC$



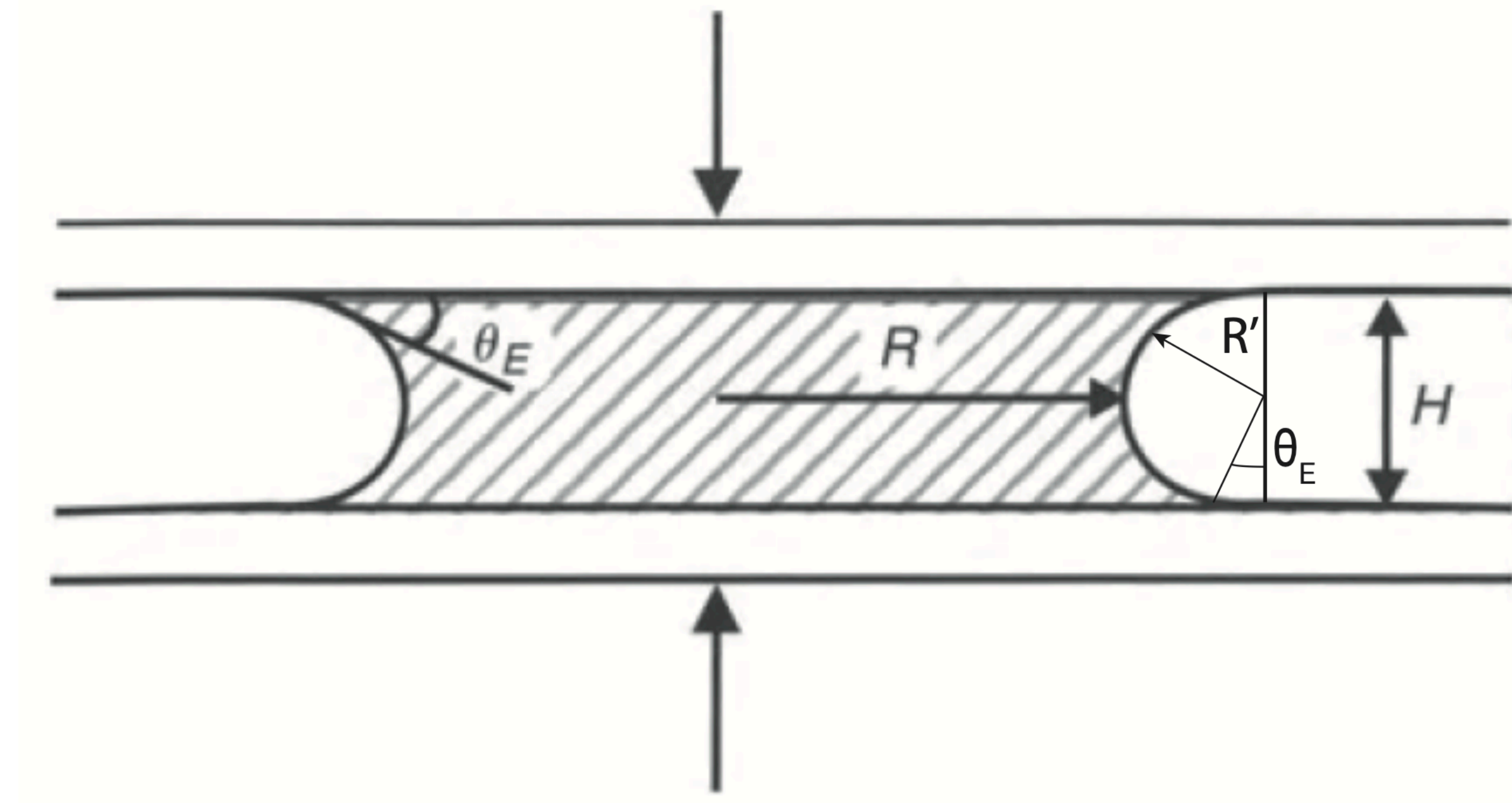
Formation des micelles :
 $c > CMC$



Forte concentration en micelles : $c \gg CMC$

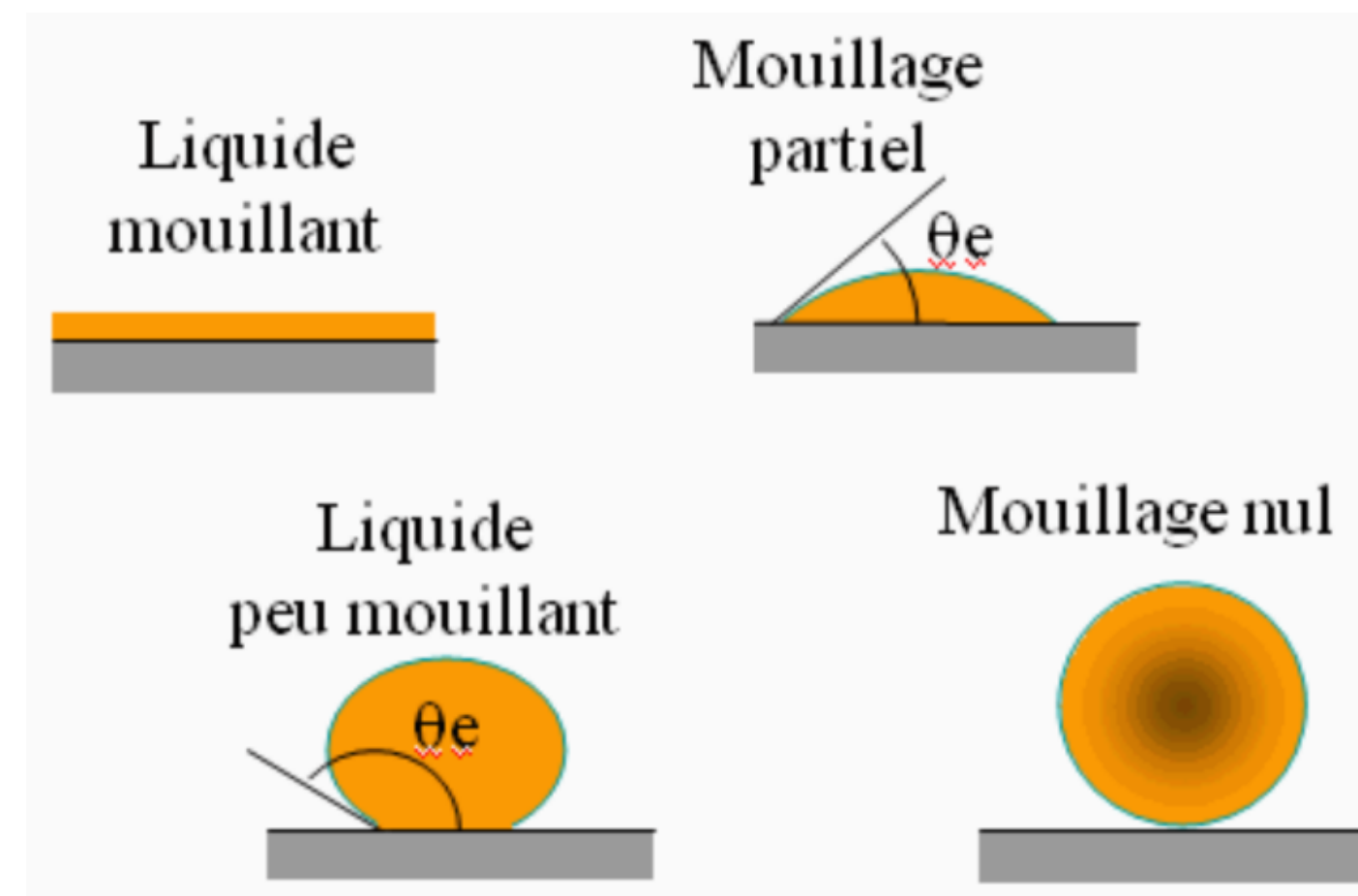
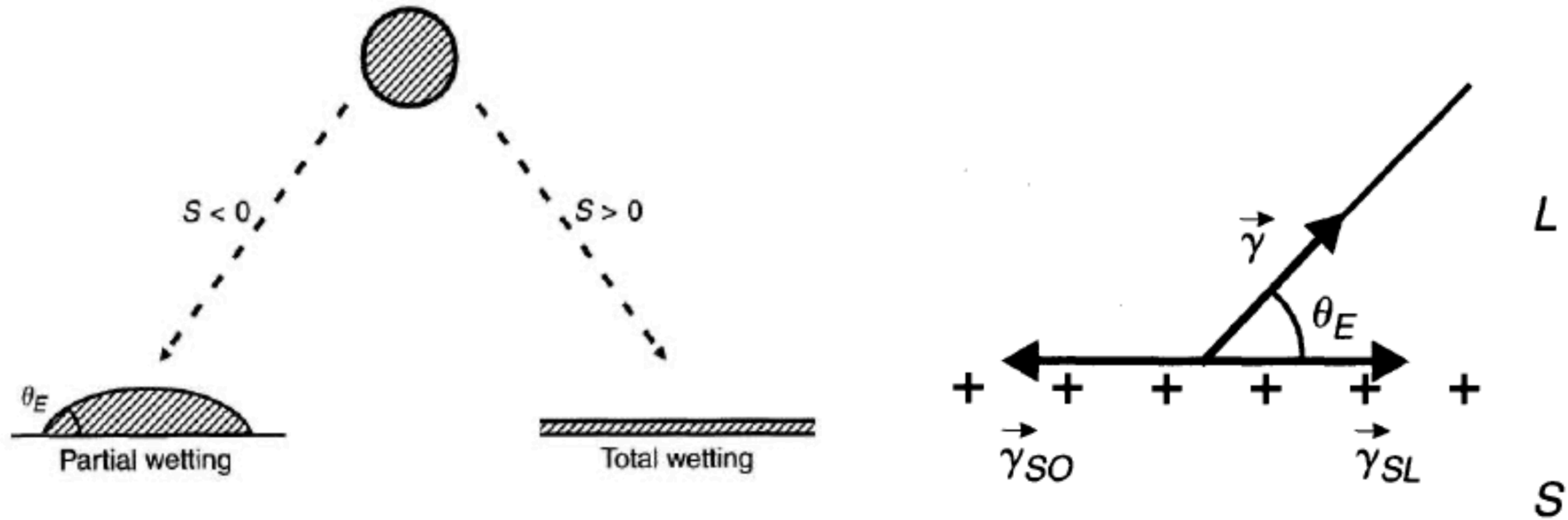


b) Application : adhésion capillaire (« pont capillaire »)

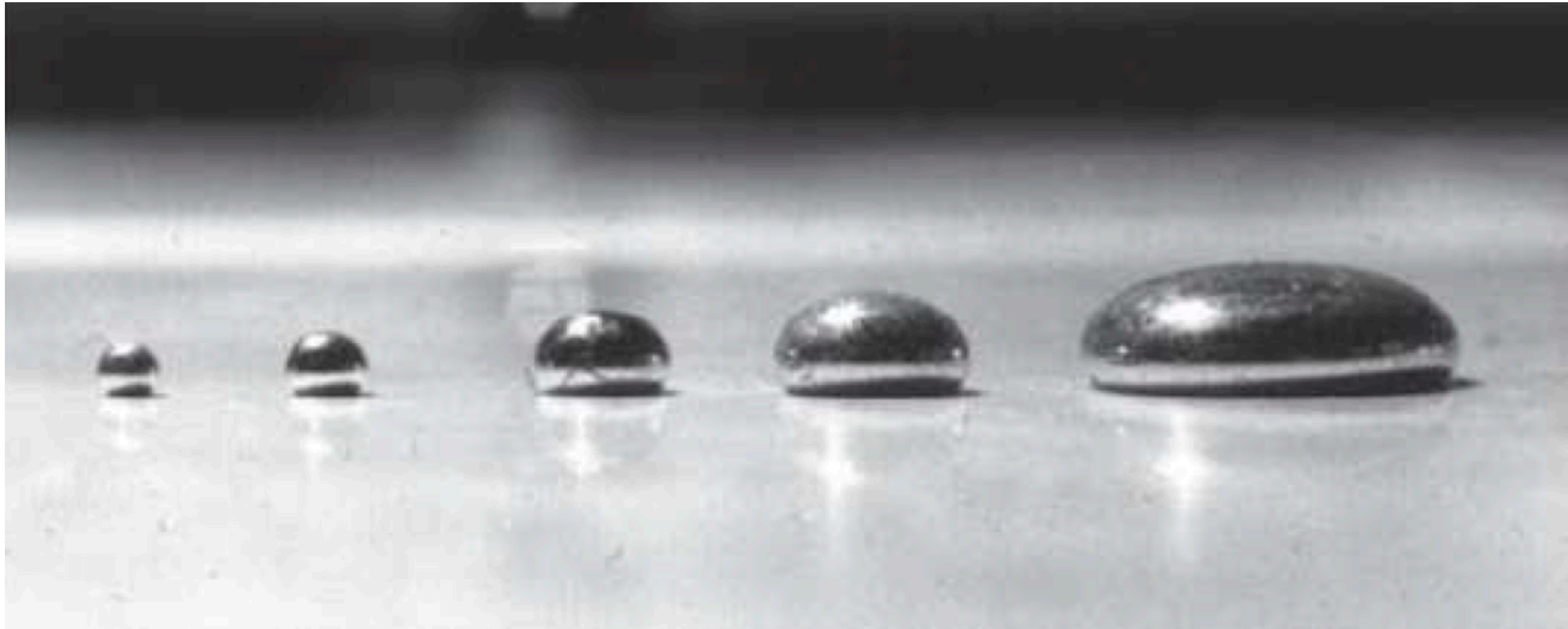


$$\Delta p \sim -\frac{2\gamma \cos(\theta_E)}{H}$$

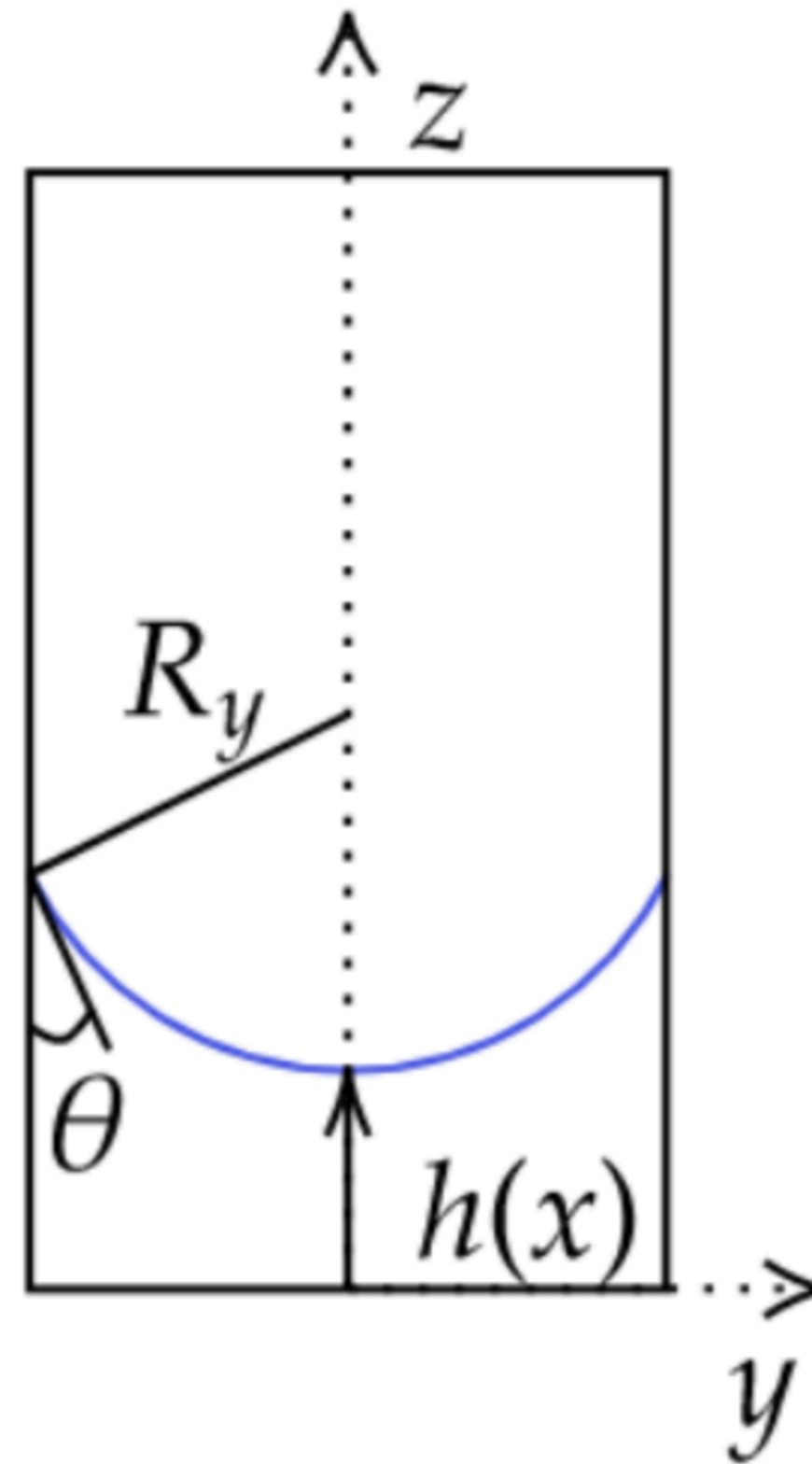
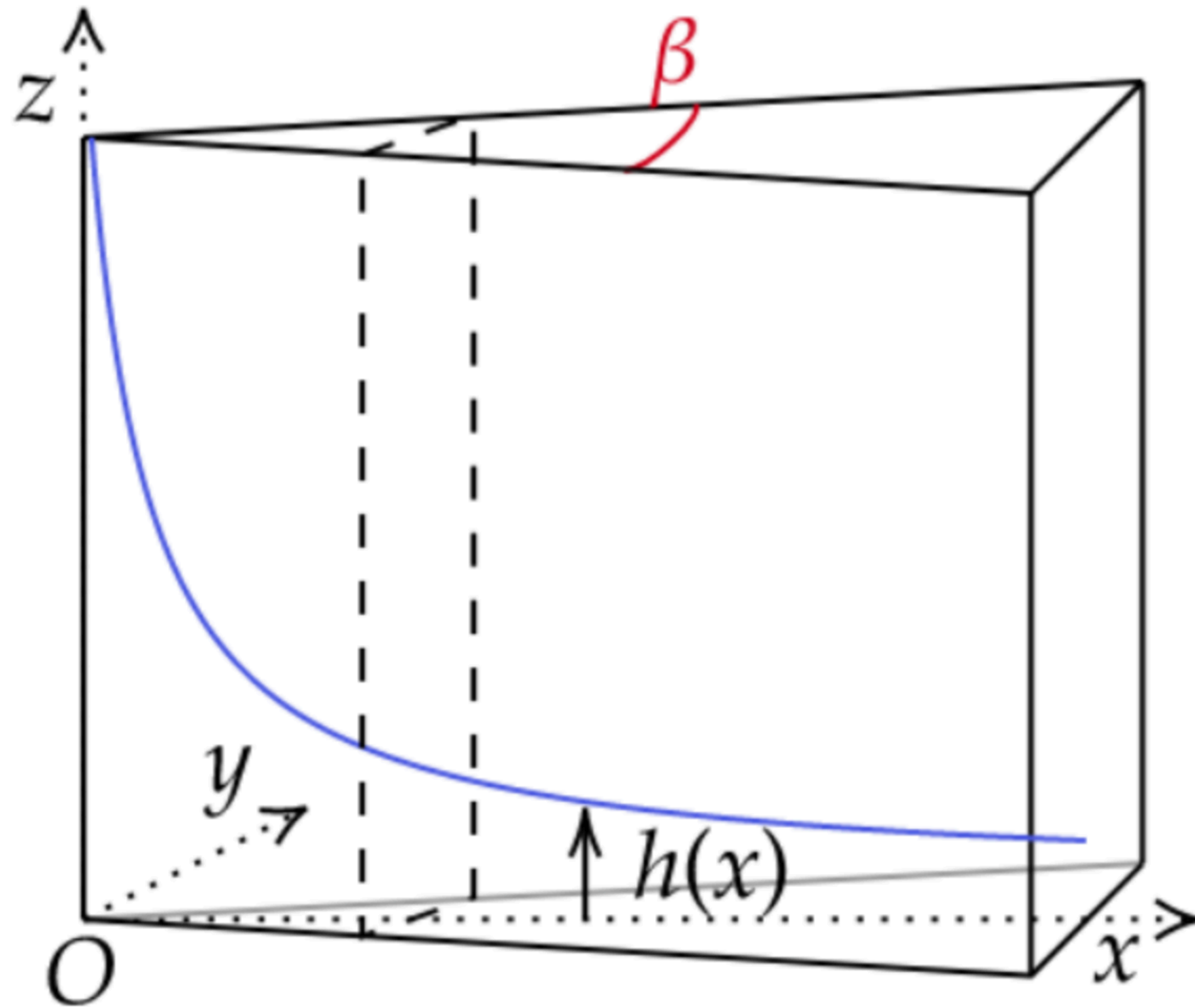
a) Angle de mouillage



b) Influence de la gravité

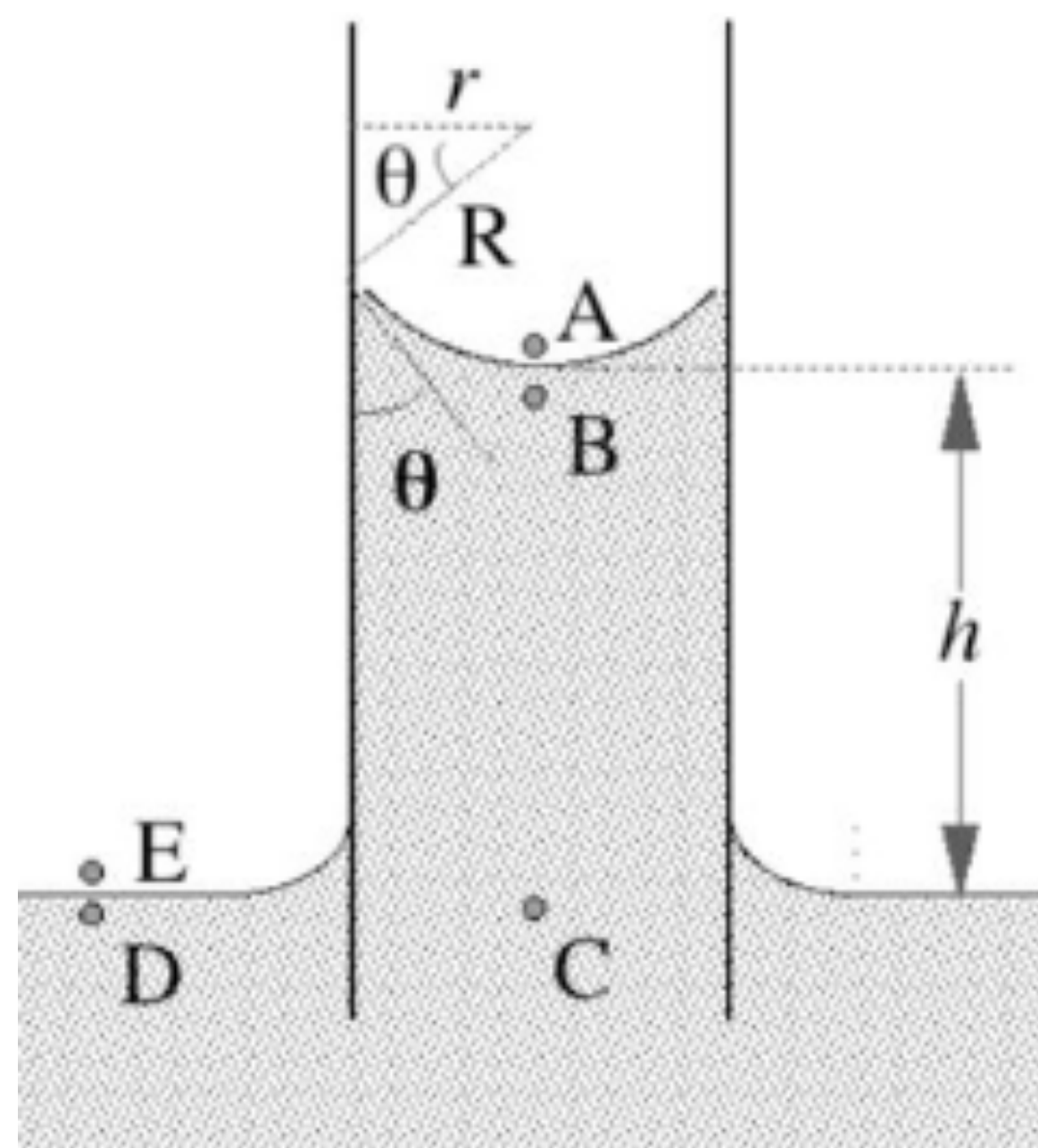


c) Ascension capillaire



$$h(x) = \frac{\ell_c^2}{x} \frac{\cos(\theta)}{\tan(\beta/2)}$$

$$lc = \sqrt{\frac{\gamma}{\rho g}}$$



$$h = \frac{2 \gamma}{\rho g} \frac{1}{R}$$